

Placement Readiness 5-Day Workshop

Master Handbook for Problem Solving, High-Frequency DSA Invariants, and Platform Administration

100 Total MCQs (10 Pre + 10 Post/Day)

25 Solved Exemplar Programs (5/Day)

10 Level 0 Syntax Foundations

11 Algorithm Visualizer Simulations

25 Placement Aptitude MCQs

10 Spinning Wheel DSA MCQs

Real Google Authentication Sync

CHIEF INSTRUCTOR

Kapil

TARGET AUDIENCE

**Placement Aspirants
& Administrators**

PLATFORM VERSION

& Production 2.0 • 2026 Edition

1. Executive Overview & Pedagogical Architecture

STRATEGY

Core Mission: Product companies (Amazon, Microsoft, Google, Meta, TCS Digital) evaluate candidates not on code memorization, but on *boundary invariance*, *space-time trade-offs*, and *conceptual debugging*. This platform operationalizes a disciplined, gated curriculum ensuring zero concept leakage.

1.1 Pedagogical Matrix: What, Why, How

DIMENSION	LEARNER PERSPECTIVE	ADMINISTRATOR PERSPECTIVE
WHAT	A rigorous 5-day bootcamp featuring 100 MCQs, 25 multi-language exemplars, 11 algorithm visualizers, Level 0 syntax foundations, built-in 6-language IDE, and daily feedback.	A centralized command portal with live Google Firebase sync (zero fake users), recursive day lock controller, CSV reporting suite, and manual credential issuance.
WHY	Prevents tutorial-hell by forcing diagnostic evaluation before coding, multi-language comparison, and passing scores ($\geq 70\%$) to unlock verified placement badges.	Provides absolute batch control, preventing learners from jumping ahead before class, while supplying college placement cells with verified daily attendance & performance audits.
HOW	Sign in with Google → Clear 10 Pre-MCQs → Acknowledge 5 Solved Exemplars → Solve Guided Tasks in IDE/HackerRank → Score $\geq 70\%$ in 10 Post-MCQs → Submit 2:30 PM Feedback.	Access portal via discrete Lock icon → Authenticate (<code>KAPILADMIN</code> / <code>ADMIN123</code>) → Inspect real learner progress → Toggle global or individual day locks → Export CSVs.

1.2 Curriculum Blueprint Across the 5 Days

DAY	CURRICULUM TOPICS	PRE-TEST	EXEMPLARS	POST-TEST	DAILY BADGE
Day 1	T1: Graphs (BFS/DFS Reachability) • T2: Strings & Recursion	10 MCQs	5 Solved	10 MCQs	GRAPH MASTER
Day 2	T3: Recursion & LCM/GCD • T4: Algorithmic Complexity & Math	10 MCQs	5 Solved	10 MCQs	MATH WIZARD
Day 3	T5: Number Theory (Sieve) • T6: Two-Pointer Techniques	10 MCQs	5 Solved	10 MCQs	POINTER GURU
Day 4	T7: Advanced Pointers (Sliding Window) • T8: Divide & Conquer	10 MCQs	5 Solved	10 MCQs	D&C EXPERT
Day 5	T9: Advanced D&C • T10: Matrix Manipulations & Synthesis	10 MCQs	5 Solved	10 MCQs	CAPSTONE CHAMP

2. The Comprehensive Learner Operating Guide

2.1 The Mandatory 4-Step Daily Learning Progression

Every workshop day is structured around a strict sequential flow. Learners cannot skip steps or unlock Day $N+1$ without fulfilling all criteria of Day N .

STEP 1

Diagnostic Pre-Test

10 MCQs evaluating fundamental math, recursion trees, and complexity bounds before touching code.

STEP 2

5 Solved Exemplars

Study 2 Easy, 2 Med, 1 Hard programs with multi-language code and click *Mark as Complete*.

STEP 3

Guided Syllabus Tasks

Solve in-class and post-class coding challenges on **HackerRank** and in the built-in **6-Lang IDE**.

STEP 4

Mastery Post-Test

Take 10 Post-MCQs based on today's algorithms. Score $\geq 70\%$ to earn your verified Badge.

2.2 Step-by-Step Learner Execution Protocol

- ✓ **Authentication:** Click *Sign In with Google* in the top header. Your cloud session ensures all progress, scores, and badges persist across devices.
- ✓ **Pre-Assessment:** In the Day view, click *Start Pre-Test*. Submit your 10 answers. This records your baseline readiness score.
- ✓ **5 Solved Practice Questions Masterclass:**
 - Switch between **C++**, **Java**, **Python**, **C**, and **HTML/JS** tabs to inspect syntactic nuances.
 - Read Kapil's Intuition notes detailing edge-case traps (e.g., `INT_MIN` boundaries in median partitions).
 - Click **Open in IDE** to test code with custom inputs.
 - Click **Mark as Complete** for all 5 problems to satisfy progression criteria.
- ✓ **Solving Guided Problems:** Tackle in-class problems directly in the integrated compiler or on HackerRank. Mark each problem as solved upon completion.
- ✓ **Earning Badges via Post-Assessment:** Complete the 10 Post-Assessment questions. Scoring 7/10 or higher immediately awards the day's badge.
- ✓ **Daily Workshop Feedback (fpIn.site):**
 - Window: **2:30 PM to 3:30 PM IST daily**.
 - Automated countdown banners notify learners at 2:00 PM (30 min reminder), 2:20 PM (10 min), and 2:25 PM (5 min).
 - Click *Open Day Feedback* to record your evaluation.
- ✓ **Unlocking Subsequent Days:** When Day 1 is 100% complete, Day 2 unlocks automatically. If Day 1 remains incomplete, Day 2 displays the secure lock gate.

2.3 Supplemental Accelerator Tools

MODULE	LOCATION	PURPOSE & PEDAGOGICAL VALUE
Level 0 Foundations	Sidebar Top / Day Tab	10 solved introductory programs in Java, C, C++, Python, and HTML/JS designed for syntax warm-up and cross-language translation practice.
Algorithm Visualizer	Sidebar / Day Tab	11 interactive simulations (Searching, Sorting, Graphs, Trees). Provides visual feedback with animated histograms and donut nodes with up to 3 test tries.
25 Aptitude MCQs	Main Home Screen	Placement-specific quantitative aptitude, speed math, permutations, probability, and work-rate reasoning.
Spinning Wheel Lab	Header / Welcome Card	10 gamified DSA challenge questions featuring realistic negative marking to train risk-calibrated test taking.

MODULE	LOCATION	PURPOSE & PEDAGOGICAL VALUE
In-Browser IDE	Problem Cards	Full-featured coding environment supporting 6 languages with live test-case runner, custom input execution, and timing metrics.

3. Super Admin Operating Guide (Kapil's Control Center)

INSTRUCTOR
PORTAL

Zero Fake Data Guarantee: The Admin Dashboard syncs directly with live Google Firebase Authentication. No mock, placeholder, or generated student records exist. Only real students with authenticated sessions are displayed.

3.1 Super Admin Authentication & Portal Access

To preserve learner immersion, admin controls are discreetly embedded behind a secure modal with zero on-screen credential exposure.

TOUCHPOINT	ACCESS LOCATION	CREDENTIALS
Desktop Web	Lock icon in top-right Header and Left Sidebar navigation.	Username: KAPILADMIN Password: ADMIN123 <i>(Inputs are masked with an optional visibility toggle)</i>
Mobile / Tablet	Floating gold Lock action button fixed in bottom-right corner.	

3.2 Core Administrative Operations & Workflows

A. Day Unlock & Relock Controller (Batch & Individual Management)

Instructor Kapil maintains absolute authority over content access. Days can be toggled globally or customized for individual learners.

- ✓ **Unlock All Days Globally:** Bypasses all locks simultaneously for the entire cohort (ideal for weekend review sessions or open mock tests).
- ✓ **Relock All Days to Default:** Restores strict chained gating (Day *N* requires Day *N-1* completion).
- ✓ **Individual Student Remediation:** Navigate to the student roster → Click *Audit & Access* → Toggle individual day overrides (**FORCE UNLOCK** or **FORCE LOCK**). Real-time events update the student's screen instantly without requiring page reloads.

B. Automated Placement Cell CSV Export Engine

College placement cells and training coordinators require clean, timestamped compliance data. The dashboard provides one-click CSV generation:

- ✓ **Daily Reports (Day 1 to 5):** Exports student names, emails, UID, Pre-Test score, Post-Test score, exemplar count, and feedback status for that specific calendar day.
- ✓ **5-Day Master Consolidated Report:** Comprehensive multi-column spreadsheet containing total questions solved, all 5 pre/post scores, total badges won, and final certificate eligibility.

C. Credentials, Badges & Curriculum Extensions

Admin Kapil can issue official credentials and extend the curriculum directly from the portal:

- ✓ **Issue Custom Badges:** Select student, select badge title (e.g. *Graph Master*, *Dynamic Programming Wizard*), add personalized commendation notes, and issue directly to their profile.
- ✓ **Issue Verified Certificates:** Generate authenticated PDF-ready certificates for students achieving capstone milestones.
- ✓ **Deploy Assignments & Quizzes:** Create new coding challenges with starter code and custom test cases, or author new multi-choice quizzes with detailed rationales.

4. Security Controls & Operational FAQ

4.1 Security Architecture & Gating Rules

Strict Gating Invariant: When Day *N* is locked, the application completely blocks all underlying problem statements, assessment questions, and starter code. No inspection or manipulation can expose locked curriculum data.

COMPONENT	LEARNER VIEW	SUPER ADMIN VIEW
Locked Day Access	Renders dedicated Inaccessible Gate with prerequisite checklist and return button.	Displays lock notice with direct <i>Unlock via Admin Panel</i> override button.
Bypass Controls	Completely removed from DOM. Learners have zero bypass buttons.	Available in Admin Dashboard and persistent Admin Session banner.
Session Persistence	Progress synced to Firebase Cloud via Google OAuth UID.	Admin session stored in encrypted local storage with instant logout.

4.2 Frequently Asked Operational Questions

SCENARIO / QUESTION	ROOT CAUSE	RESOLUTION PROCEDURE
Day 2 remains locked even though learner solved all Day 1 coding problems.	Missing Pre-test, Post-test, or exemplar acknowledgment.	Learner must verify: (1) Day 1 Pre-test completed, (2) all 5 Solved Practice Questions marked as Complete, and (3) Post-test scored $\geq 70\%$. Alternatively, Admin Kapil can force-unlock Day 2 for that student.
Learner was absent for Day 1 but needs to attend Day 2 live session.	Sequential prerequisite dependency block.	Admin Kapil opens Admin Dashboard → Finds student email in Roster → Clicks <i>Audit & Access</i> → Toggles Day 2 to FORCE UNLOCK .
Learner claims feedback form link is not working.	Accessing outside scheduled evaluation window.	Feedback form (<code>fp1n.site</code>) operates between 2:30 PM and 3:30 PM IST daily. Outside this window, submission is locked by the server.
Where can placement coordinators retrieve attendance data?	Reporting requirement.	Admin Kapil accesses <i>CSV Reports Center</i> tab → Clicks <i>Download Master 5-Day CSV</i> → Shares spreadsheet with placement cell.